Gateways, BLE Clients and Asynchronous Notifications

Communication Technologies for the Internet of Things, Lesson 06

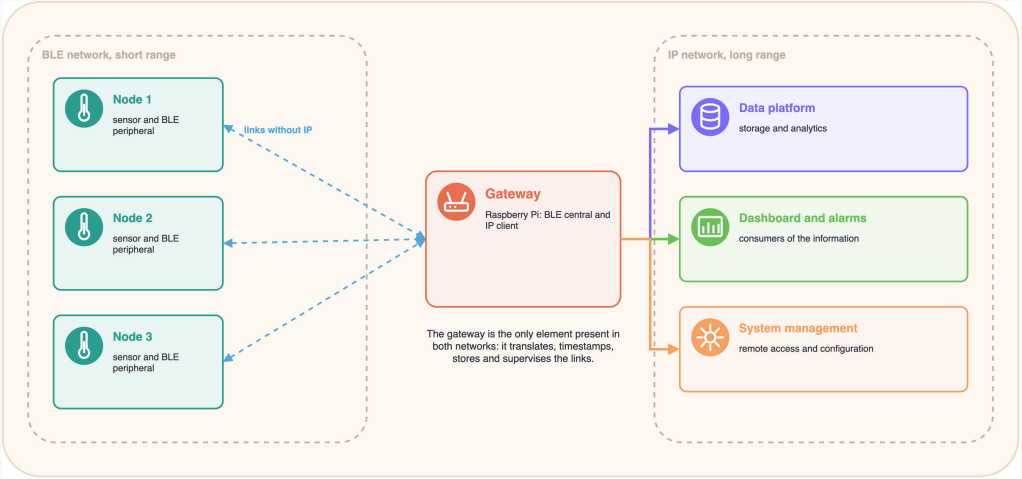
Catarina Silva

ESTGA, University of Aveiro | 2026

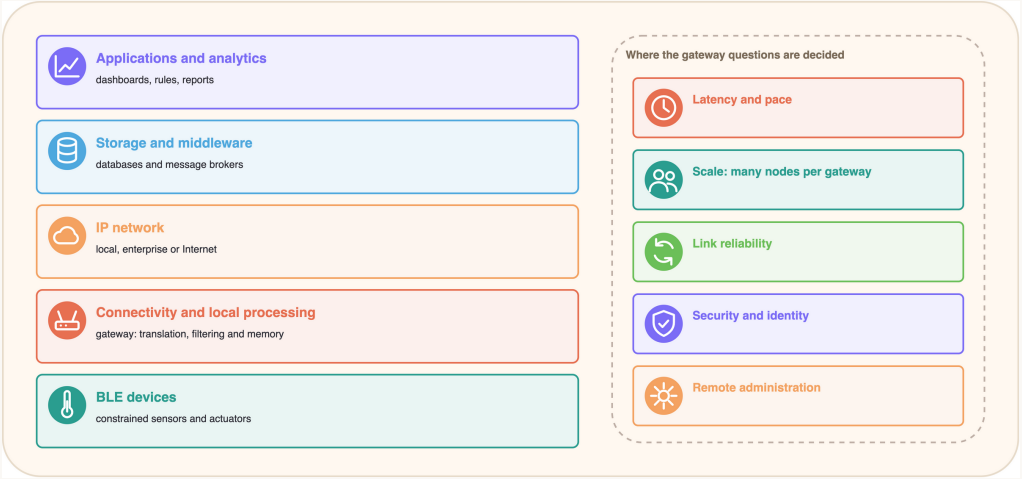
Agenda

- The architecture of an IoT system with a BLE gateway
- How a BLE session works and what the client sees
- Latency, throughput and link reliability
- Several nodes, local processing and the IP network
- Remote administration, security and architecture requirements

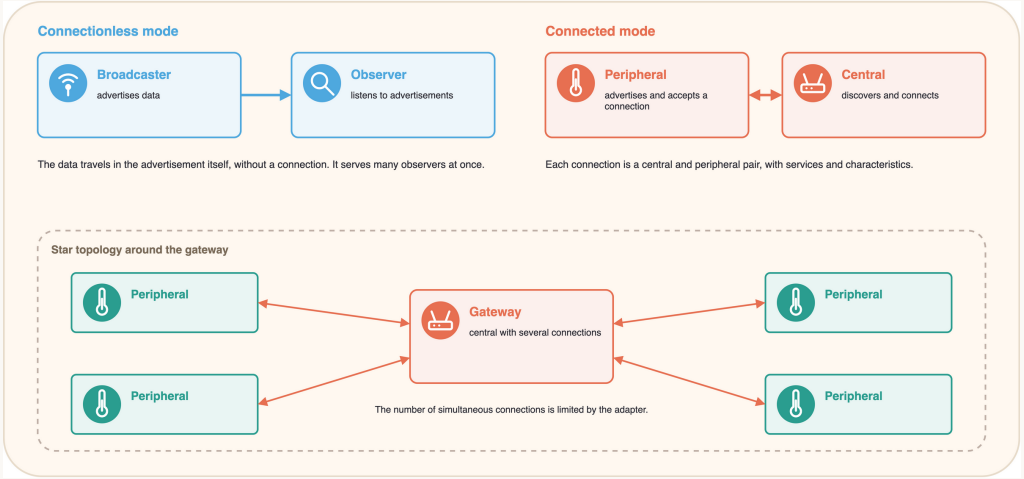
The Gateway Between Two Networks



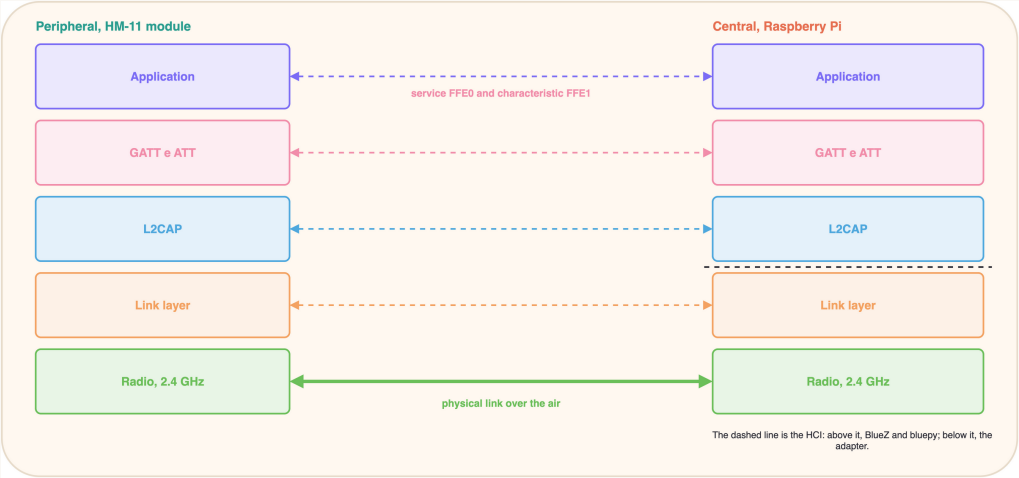
Architecture Layers



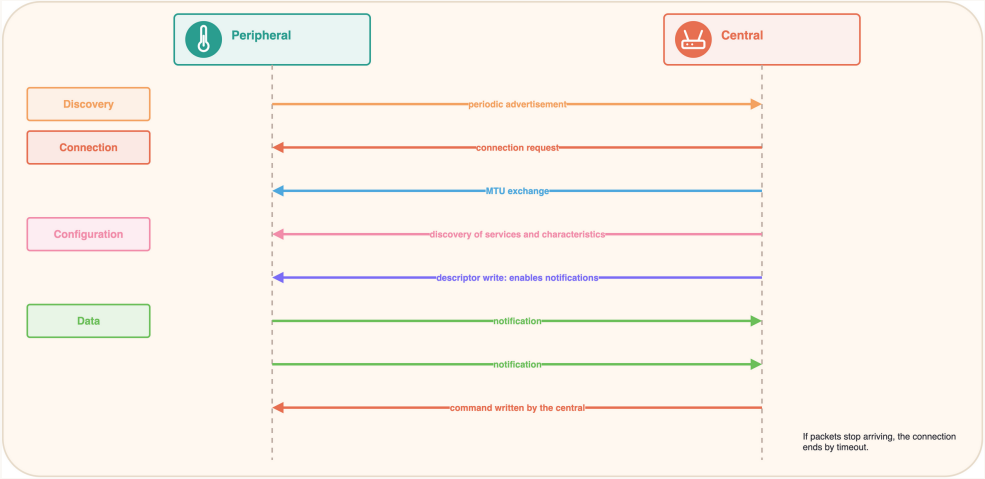
Roles and Topology



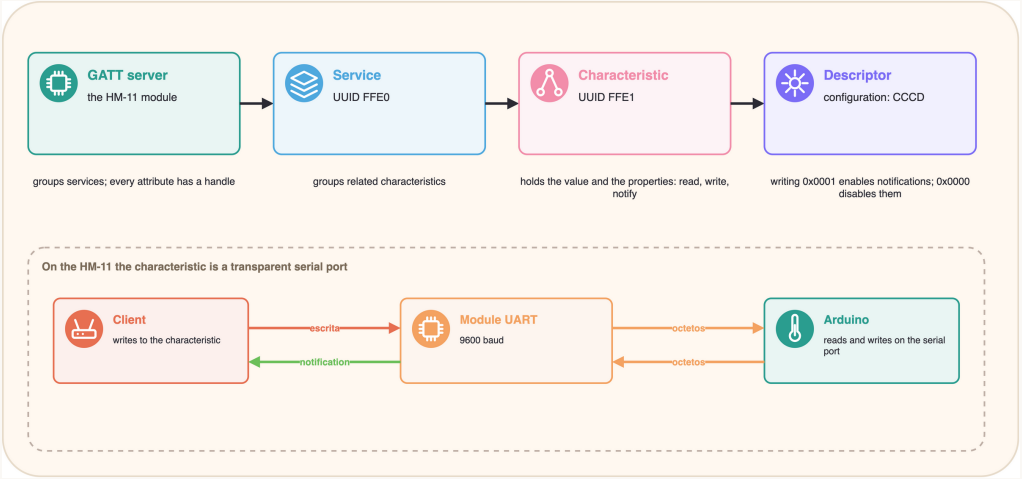
The BLE Stack at Each End



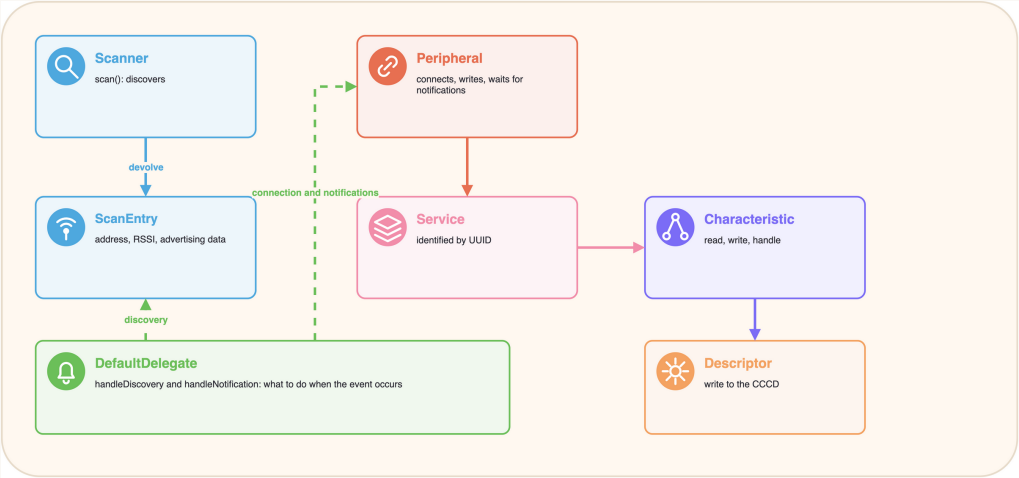
A BLE Session, from Advertising to Data



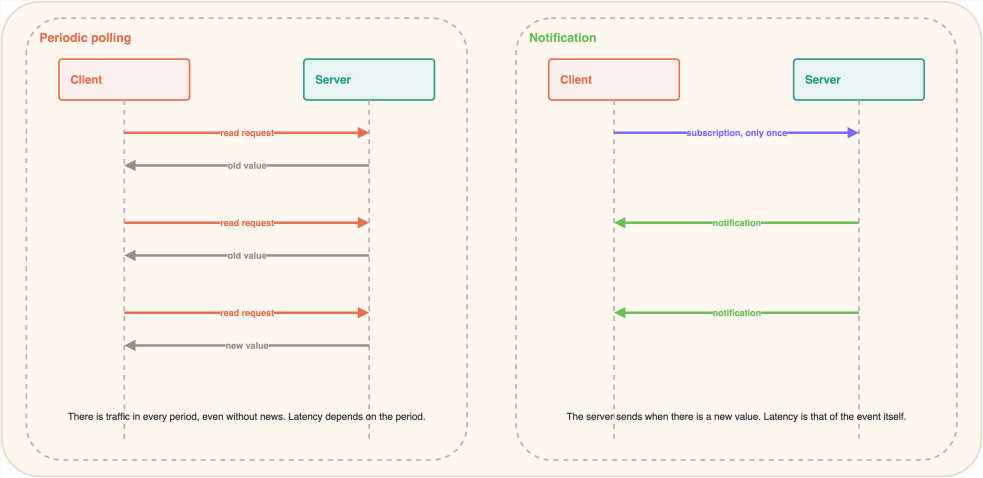
GATT Data Model



The bluepy Class Model



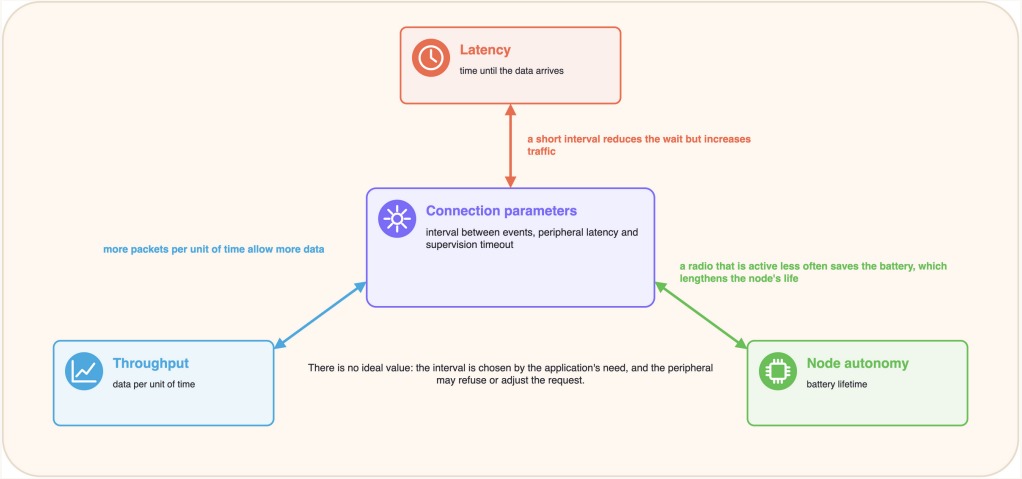
Polling and Notification



Where Latency Comes From

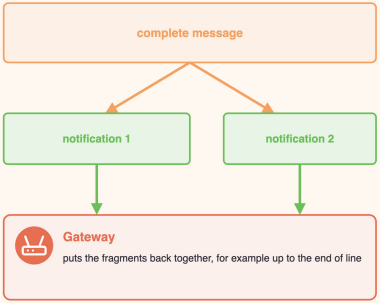


Connection Parameters and Trade-offs

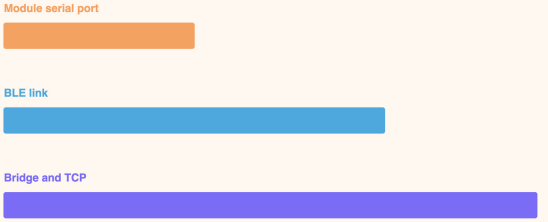


Throughput and Fragmentation

A message larger than the payload is fragmented



Useful throughput is limited by the narrowest link

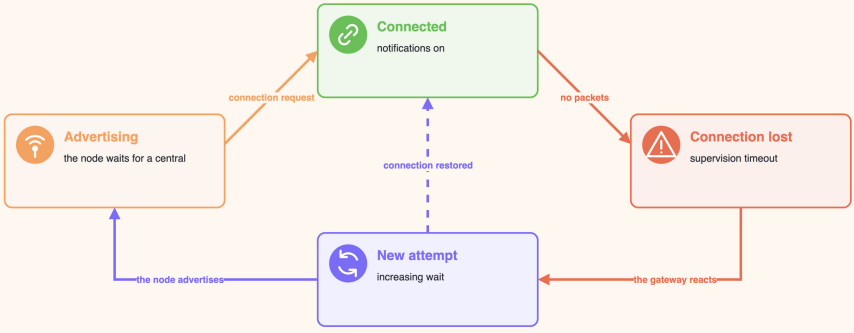


In the exercise the narrowest link is the module serial port, not BLE.

Consequence

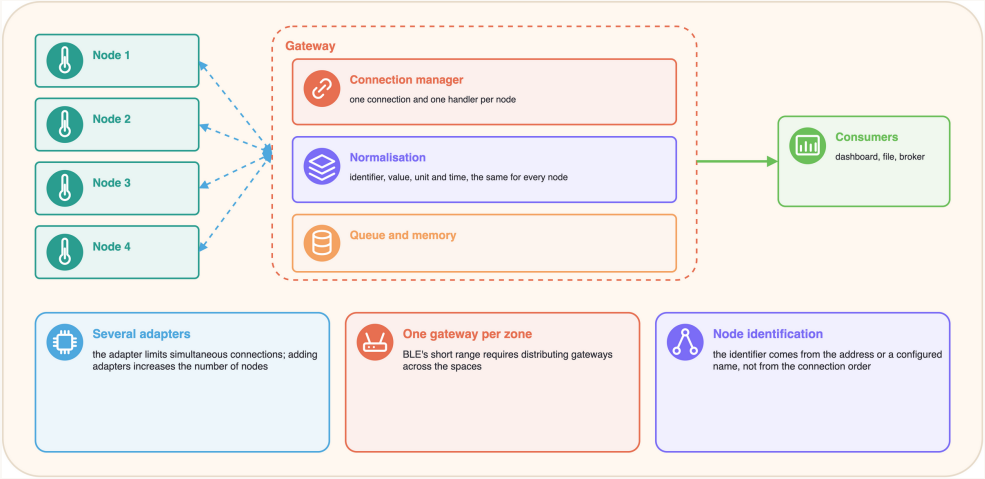
the gateway gains nothing by requesting readings faster than the node can deliver them

Link Reliability

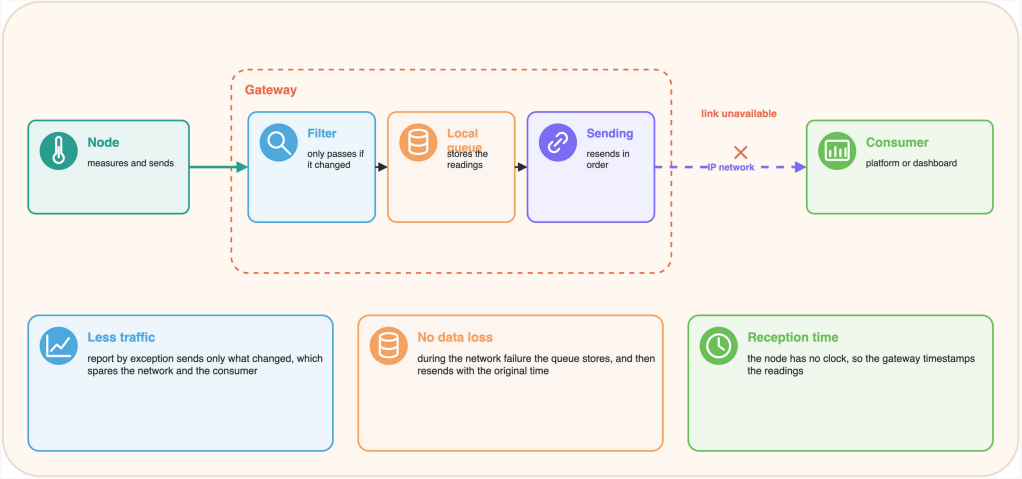


The counter in the messages measures what was lost during the failure. The log file keeps the time of each transition.

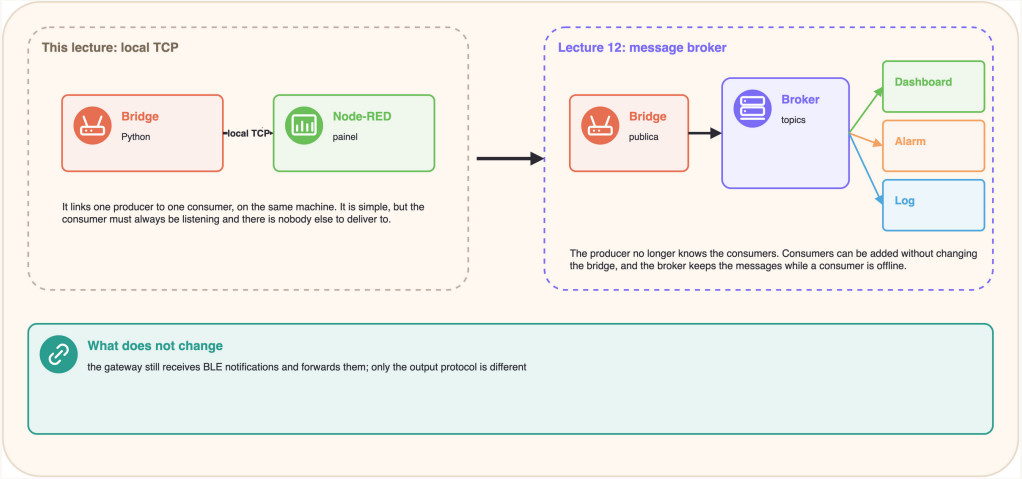
Several Nodes on One Gateway



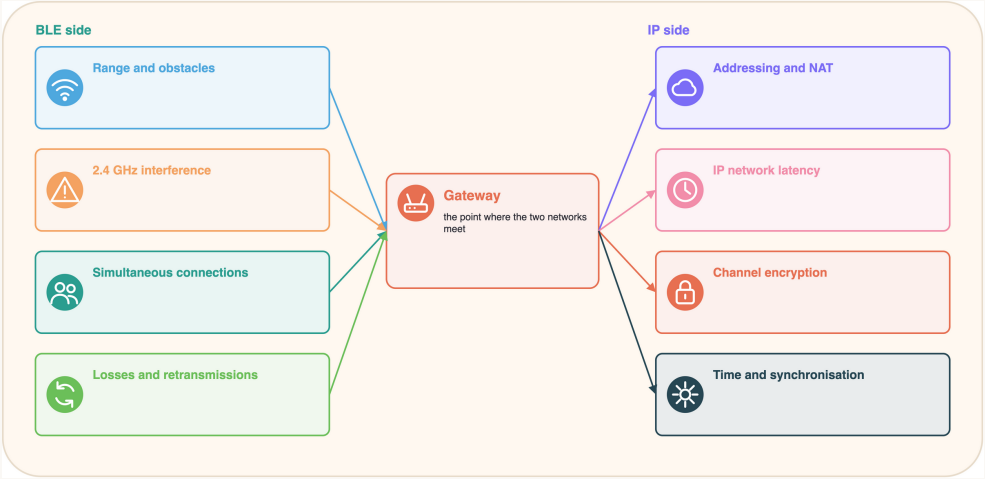
Local Processing and Memory



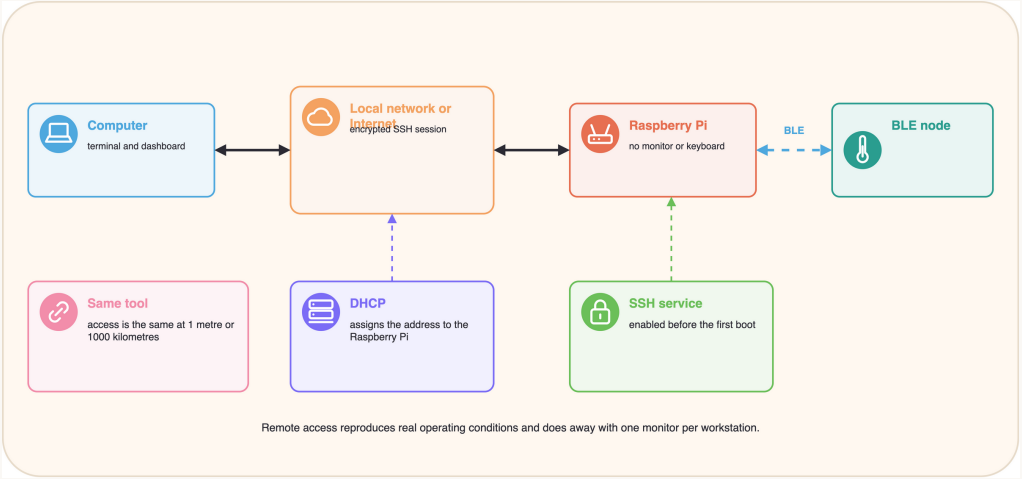
Forwarding to the IP Network



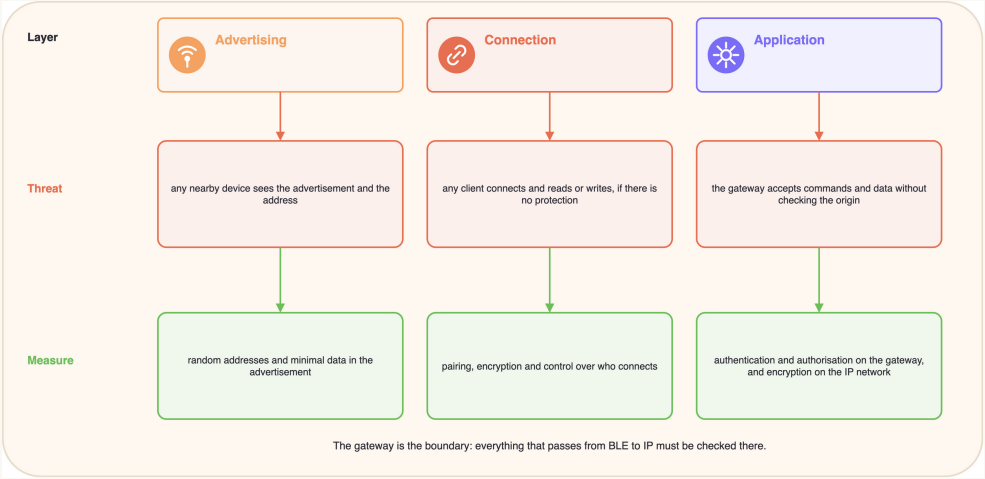
Network Questions



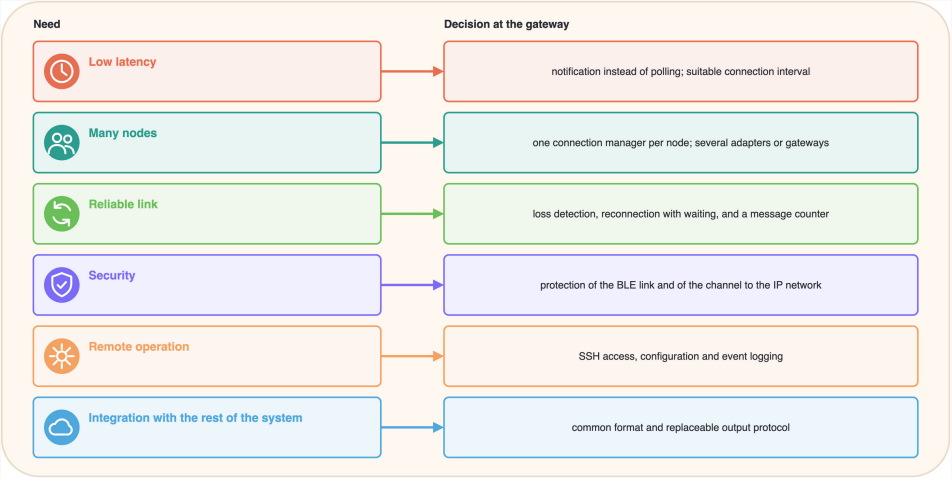
Remote Administration



Security and Privacy



Architecture Requirements



Lesson Summary

- The gateway is the only element present in both networks: it translates, timestamps, stores and supervises the links.
- Notifications remove the need for polling and reduce latency, but the final latency depends on the whole chain, not only on BLE.
- The number of nodes, the range and the reliability decide how many gateways and adapters are needed.
- Security and remote administration are part of the architecture from the start, not a later addition.

Sources

Bluetooth SIG. *Bluetooth Core Specification*, volume on the link layer, the ATT and the GATT.

Cisco. *The Internet of Things Reference Model* (2014). Positioning of the gateway at the connectivity level. Available in `Aula 01/Leituras/`.

bluepy library documentation: `ianharvey.github.io/bluepy-doc/`

BlueZ documentation: `bluez.org`

Raspberry Pi Foundation. Documentation on *headless* configuration and SSH access.

Thank you.

Questions and comments